CT861 Assignment 2

Network Protocol Analysis

General

The objectives of this assignment are as follows:

1. to get acquainted with Wireshark (current version 3.4.7).
2. to get a better understanding of TCP/IP based communications protocols.

Wireshark is an open-source network packet analyzing software, which you should download and install from http://www.wireshark.org/ on your own computer. It runs on Windows, Linux and MacOS.
Before starting the assignment, get acquainted with Wireshark by running a few capture sequences and completing the following tasks:

Marking Packets in Wireshark

Note that you can mark packets of particular interest. This is useful for printing (see below). Marked packets stand out with a black background and white text. To mark a packet, right-click it in the "Packet List" pane and choose **Mark / Unmark Packet** from the pop-up. To unmark a packet, choose **Mark / Unmark Packet** from the pop-up again. You may mark as many packets as you wish in a capture.

Printing Packets and Making Screenshots

To print a packet, use **File** => **Print**, choose **Selected packet only** or **Marked Packets**, choose **Packet summary line**, and select the minimum amount of packet details that you need to answer the question.

To make a screen shot of the active window, press the **Alt + Print Screen** (PrtScr) keys on the keyboard (you will have to look for these keys on your keyboard, since its placement varies with the type of keyboard). You can then paste the picture into your report or save it as a file.

Part – A: Protocol Stack Analysis

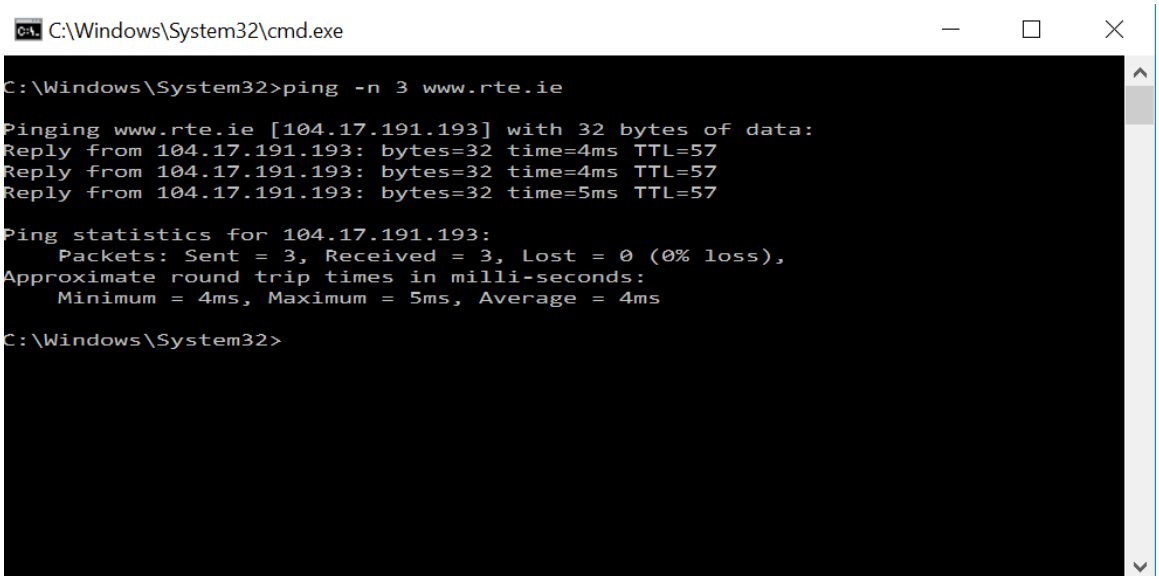
You are going to investigate the TCP/IP protocol stack running on your computer by analyzing the packets generated when you:

- 'ping' a remote machine
- open a web browser and access a website.

1. Go to the command prompt on your machine (type **cmd** from "Start➔Run" on a Windows machine, find a **Terminal** app on a Mac, and if you have a Linux OS, you probably know this already!)
   Type in **ping –n 3 www.rte.ie (or ping –c 3 www.rte.ie)**

   (See below for Windows output).
   This sends 3 ICMP ping packets to www.rte.ie and analyses the response times (min/max/avg) and lost packets. RTE's IP address may be 4 bytes (IPv4, as shown in the screenshot below) of 16 bytes (IPv6) long.
   Now start up the Wireshark packet sniffer and begin a packet capture. Repeat the ping command as above and stop capture.
   Note the extent of packets that are captured i.e. more than 6 packets!! (3 ping response + 3 reply). Go to "Statistics➔Capture File Properties" for summary information of capture.

   Save your capture file as *ping1*.

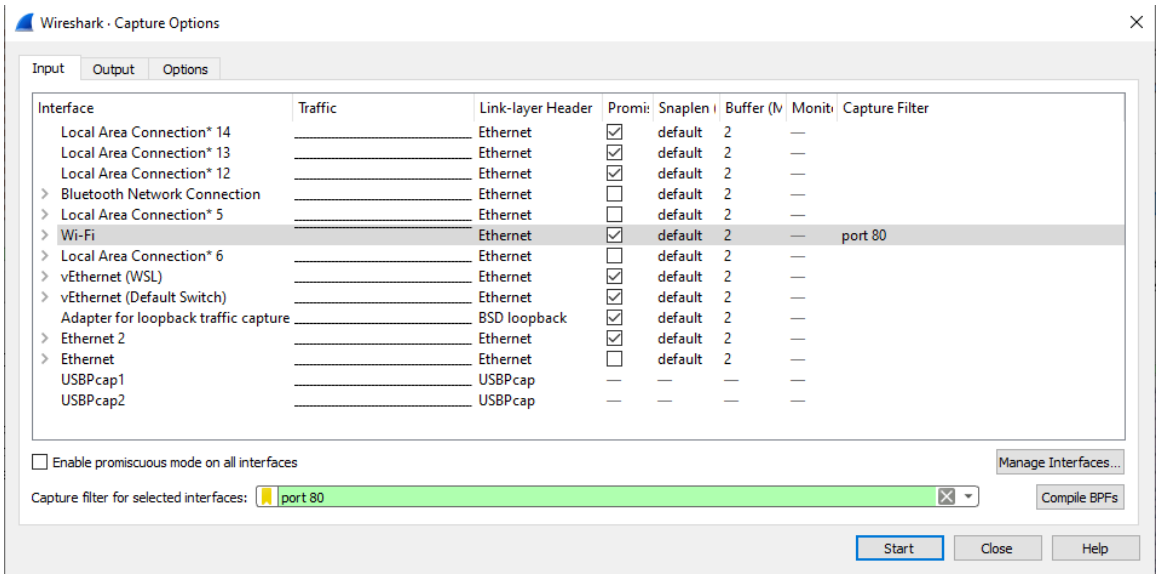


2. Now start a new capture, then open a web browser, and type in the URL of some website without HTTPS encryption, e.g. www.presathenry.ie/

   When the webpage is fully loaded stop the packet capture in Wireshark.

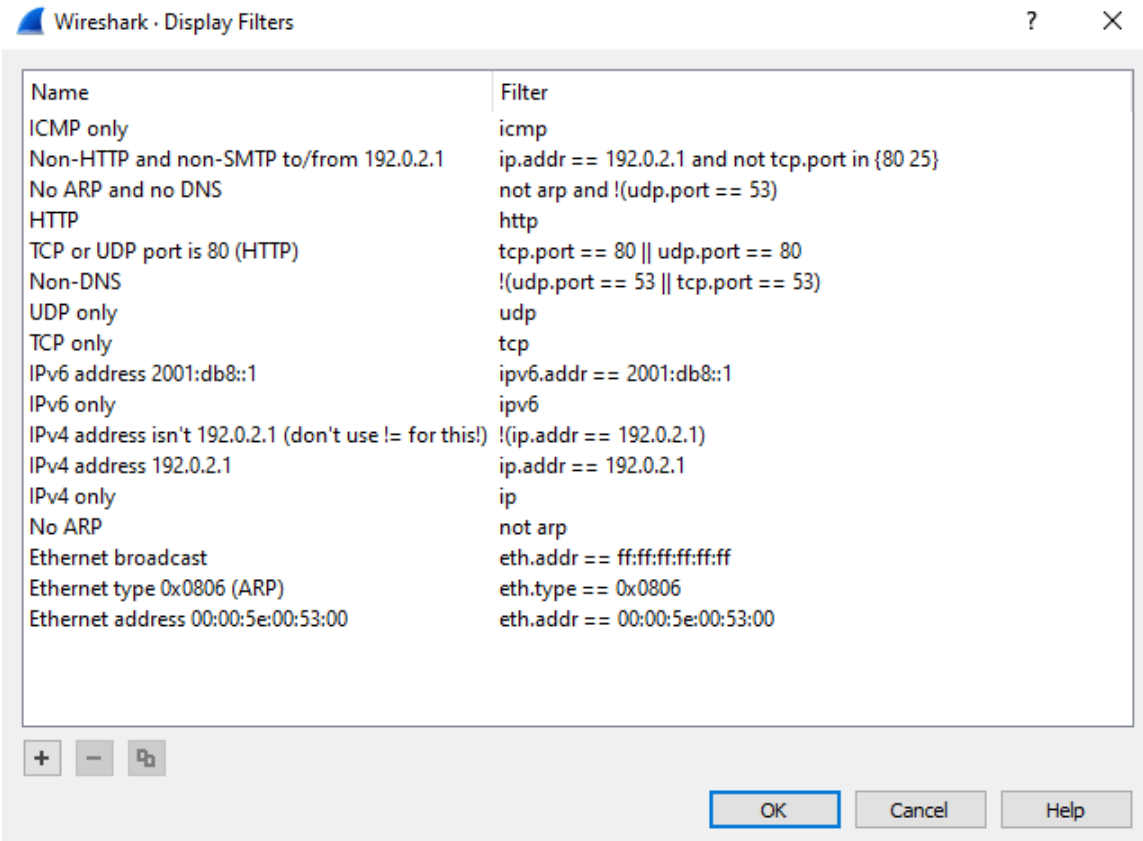
Again, save your capture as *capture1*.

Since you are focusing on specific traffic in this experiment, you can greatly simplify our subsequent analysis by using filters. Wireshark offers two types of filters: *capture filters* and *display filters*. *Capture filters* are used during the actual packet capturing process and are applied by WinPcap on windows machines and similar on other OSs. You can configure them in the **Wireshark: Capture-> Options** window where you can specify which traffic you want or do not want to be captured. For example:

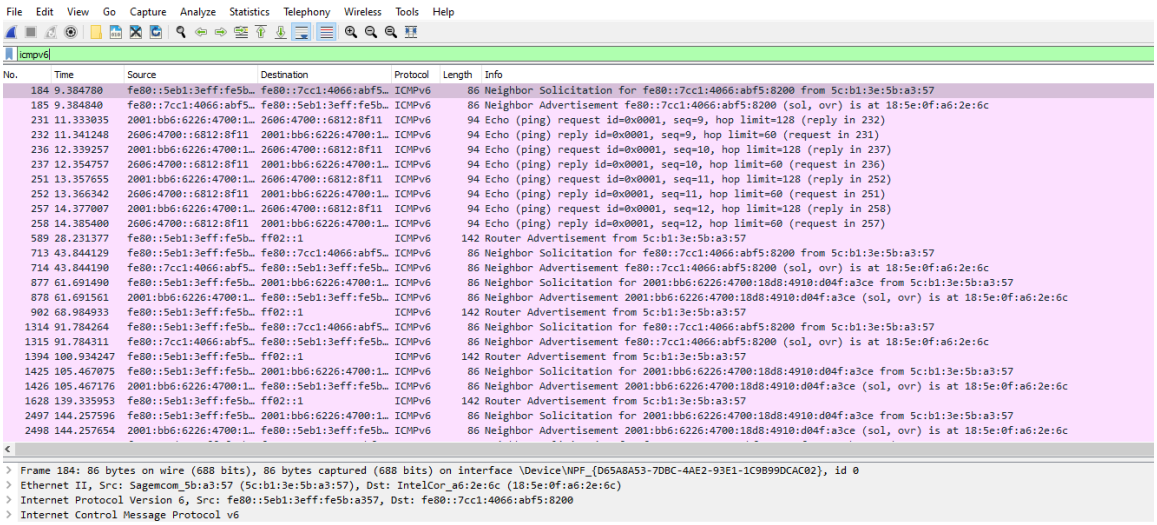


With the above filter only packets related to transport layer port 80 will be captured, and all other packets are discarded.

A 2nd option, more versatile option is to use *Display filters* applied to a capture file once that file has been created. A display filter allows you to display only packets that match that filter. You can choose or enter a display filter in the **Filter** text box ("**Analyse->Display Filters"**) as shown below.



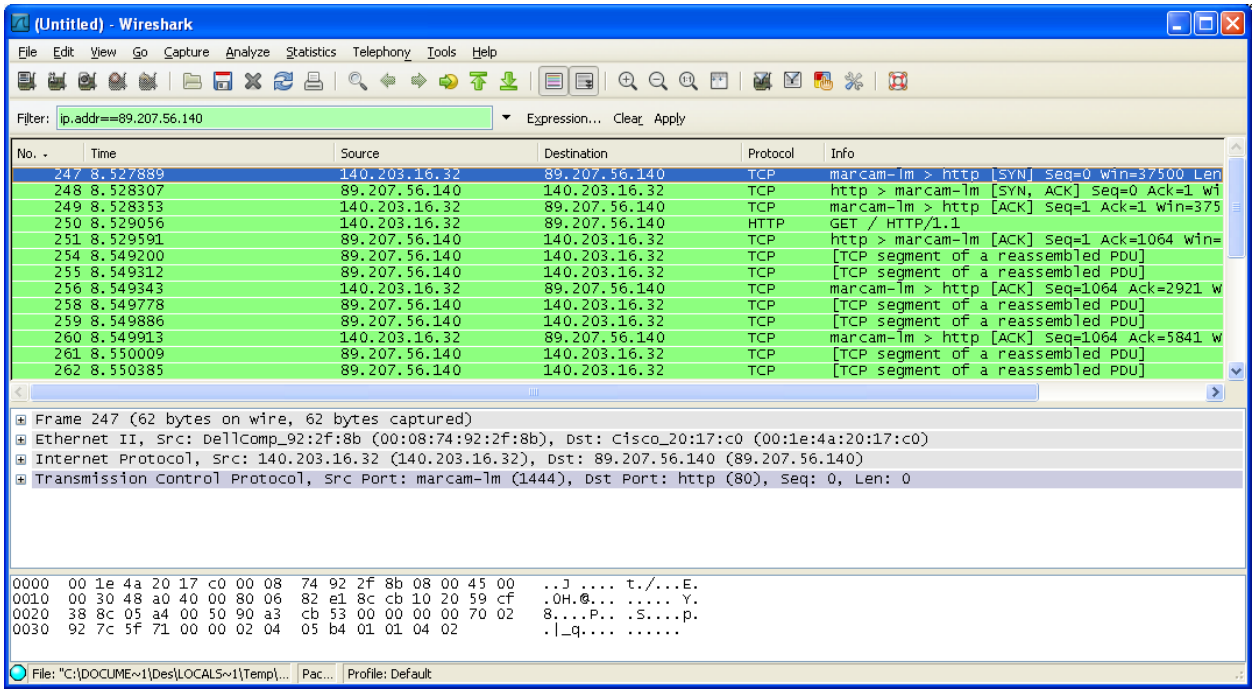
Open the saved ping1 file. In the display filter textbox, enter ‘icmpv6’ and press ‘Apply’. Only the ICMP packets are displayed from the captured file.



Display filters are more commonly used than capture filters because they allow you to filter packets without actually omitting the rest of the data in the capture file. Thus, if you

need to revert to the original capture, you can simply clear the filter expression by pressing the **Clear** button or deleting the filter string.

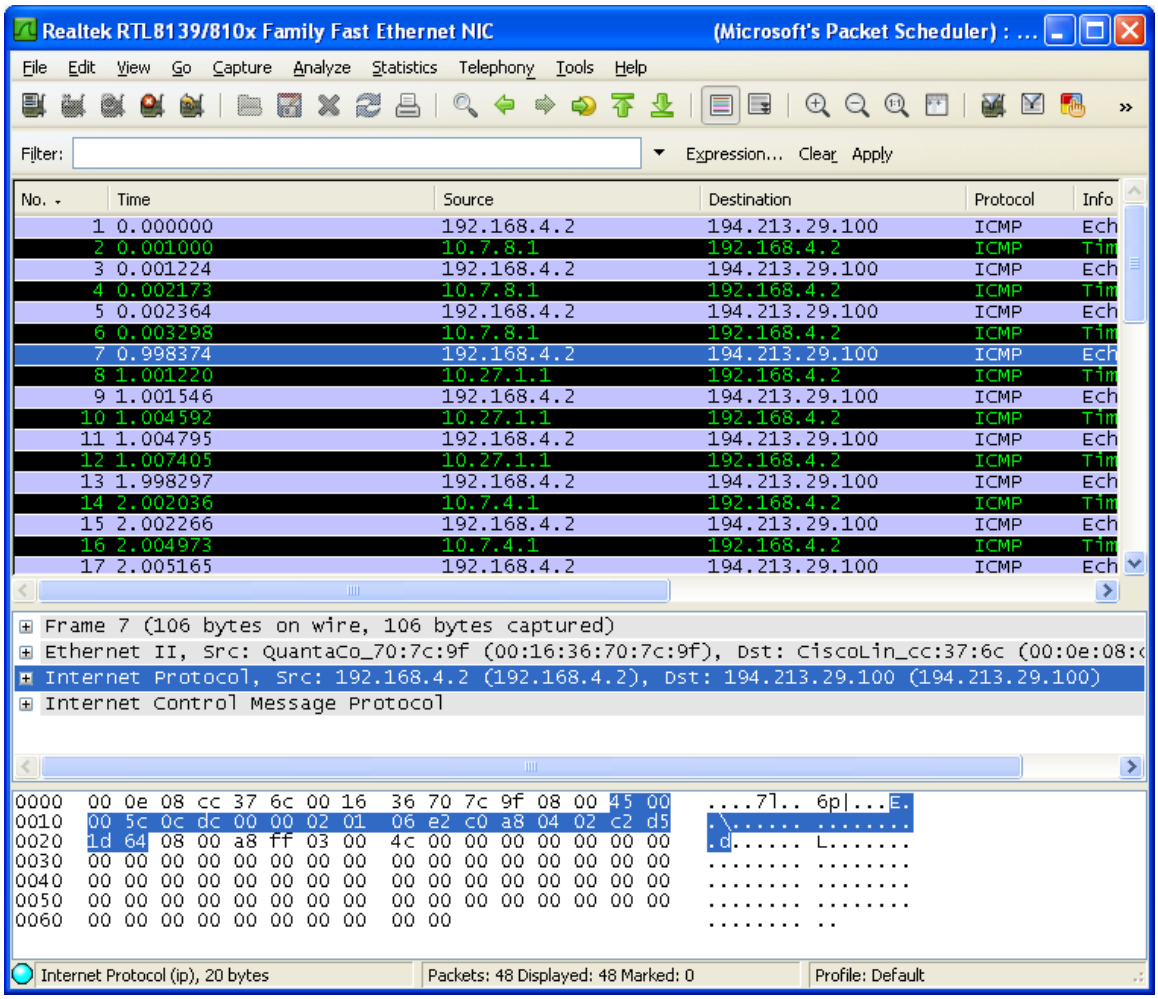
3. Open up the *capture1* file.



Note the usage of the display filter above **ip.addr==89.207.56.140** to only display traffic to and from the IP address associated with the server you have contacted - this makes it much easier to view and analyse only those packets that are of particular interest.
Further on, note that this IP address may change while you work on the assignment, and that the IP address might even change during your experiments (e.g. from 104.16.43.72 to 104.16.42.72). If this happens during your experiments, change the filter using "/", e.g **ip.addr==104.16.0.0/16** for the example above, so that only the 2 constant leftmost numbers "**104.16**" are considered by the filter.

In order to answer the questions in part B, you will need to look into the "Packet List", "Packet Details", and "Packet Bytes" panes. Ensure that all 3 windows are selected from the 'View' Toolbar option. The "Packet List" pane displays a listing of captured packets, possibly filtered by a display filter. The "Packet Details" pane shows the protocols and protocol fields of the packet selected in the "Packet List" pane. The protocols and fields of the packet are displayed using a tree, which can be expanded and collapsed. The "Packet Bytes" pane displays the data from the packet selected in the "Packet List" pane, and highlights the field selected in the "Packet Details" pane.

As usual for a hex dump, the left side of the "Packet Bytes" pane shows the offset in the packet data, in the middle the packet data are shown in hexadecimal representation (each byte is represented as a two-digit hexadecimal number), and on the right the corresponding ASCII characters (or as periods (.) if not appropriate) are displayed.



The packet capture shown on the previous page is associated with an ICMP traceroute and an ICMP message is carried inside of an IP packet, which is carried inside of a Data Link frame (e.g., an Ethernet frame in this case- as you know, Ethernet is the most common data link layer technology).

The figure below shows an Ethernet II frame and how it relates to an ICMP message.
Note: Usually, Wireshark cannot capture the FCS (Frame Check Sequence) of a frame (see the Wireshark FAQ for details). Thus, as a rule, there are no FCS bytes in captured frames.

Destination address (6 bytes)	Source address (6 bytes)	Type (2 bytes)	IP header	ICMP header	Data	FCS (4 bytes)

Ethernet II frame (payload from 46 up to 1500 bytes)

Part B - Report

Part 1: Ping [3 Marks]

1. Print the messages related Part – A 1. To start, apply the appropriate display filter to only display traffic to and from the host you accessed. Then mark all the filtered packets using the **Edit / Mark All Packets** option. Then select **Print** from the **File** menu. In the **Printer** field, specify that printing should be done in plain text using '*Output to file'*, using the filename entered in the field. In the **Packet Range** field, specify that all marked packets should be printed. In the **Packet Format** field, select the **Packet summary line** option only (i.e. uncheck the other boxes). Click **Print**. Paste the text from the output file into your report. [1 Mark]

2. Identify the full Protocol Stack involved in the **ping** exercise based on your analysis of the filtered messages. Specifically, identify the Network and Data Link protocols that were associated with the filtered messages. Distinguish between the network layer (NL) IP destination address and the data link layer (MAC) address. [2 Marks]

Report Part 2: Web Server [12 Marks]

1. Print the messages related to accessing your chosen website. As above, apply the appropriate display filter to only display traffic to and from the website you accessed (you may use the ping command again to determine the server's IP address). Then mark all the filtered packets using the **Edit / Mark All Packets** option and print as in part 1 above. Paste the text from the output file into your report**, but limit the overall amount of data to approximately 2 pages or 100 packets.** [1 Mark]

2. As above, identify the Protocol Stack running on your computer based on your analysis of the filtered messages. Specifically, identify and name the Application, Transport, Network and Data Link protocols that were associated with the filtered messages. [1 Marks]

3. Print out and explain the TCP and IP Headers for one of the filtered messages. How much space do these headers take up in the full data frame? What are the main functions carried out by these headers? What does the content of the various header fields tell you? [7 Marks]

4. Explore two analysis options in the **Statistics** menu, append their output to your report using a screen capture of the display window, and comment of the findings. [3 Marks]

Report Submission

Please submit a zipped folder to Blackboard containing:

- your report (with a cover page containing your name and student number); the report should include screen shots and printouts, and analysis.
- all capture files (ping1 etc.) from Part A.

Marking Scheme

Part 1: Ping [3 Marks]

1. Print the messages related Part – A 1…
   - 1 mark for all ping-related messages **only**, 0.5 marks otherwise

2. Identify the full Protocol Stack…
   - 1 mark for identifying the full protocol stack, i.e. Network and DLL protocols used
   - 0.5 marks each for identifying IP and MAC address

Report Part 2: Web Server [12 Marks]

1. Print the messages related to accessing your chosen website…
   - 1 mark for all messages linked to website traffic **only**, 0.5 marks otherwise

2. As above, identify the Protocol Stack …
   - 1 mark for identifying the full protocol stack, i.e. Application, Transport, Network and DLL protocols that are associated with the filtered messages

3. Print out and explain the TCP and IP Headers for one of the filtered messages…
   - IP Header (Note that depending on your setup your computer may use IPv4 or IPv6)
     - 0.5 marks for correct header length
     - 1 mark for IP source / destination and explanation / functionality
     - 1 mark for routing / hop limit (IPv6) / TTL (IPv4) / functionality
   - TCP header
     - 0.5 marks for correct TCP header length
     - 1 mark for TCP source / destination port / functionality
     - 1 mark for sequence / acknowledgement number / functionality
     - 1 mark for Windows size / functionality
     - 1 mark for header bits / functionality

4. Explore two analysis options in the **Statistics** menu…
   - 1.5 marks for each statistic, 0.5 marks for screenshot, 1 mark for comprehensive commenting